Housing First or Treatment First?

A stylized model, and the quantitative model behind it

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1 A stylized model

1.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder (SUD), and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and roughly forty parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help understand the main drivers in the model, but where my three main questions are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They face the true hazard of acquiring a SUD, they know what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, on myopia, or on a taste that changes once the household has used.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs here are a *luxury* good bought at a price: the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so left to itself the environment concentrates use at the *top*. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, never. Housing First always raises welfare when agents are forward looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD and homelessness generate externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred: children of a parent with a SUD are themselves two to four times as likely to develop one (McGovern et al., 2023), and moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 per cent (Chetty, Hendren and Katz, 2016). The model's work here is to show why the first claim holds and then to qualitatively price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

1.2 The environment

1.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are

compounded to that length: the discount factor is β per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and ϕ_0 is survival over thirty years. Their values are in Table 1 and are not repeated here. Notation follows the quantitative model throughout.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption c, housing services h and drug consumption v,

$$u(c, h, v) \;=\; (1 - \gamma - \theta)\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share. Drugs enter only in the first period, since that is the only period in which v is chosen, and housing is chosen each period from a three point grid $\{h_0, h_S, h_L\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b + v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

1.2.2 Timing

Period 1. Every household is clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability u_0, and a household that is income eligible draws a voucher with probability π_V; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r, which is normalised to one, on whatever unit it picks. It starts with no assets. Using drugs generates the risk of a SUD, increasing in v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

The transition. The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \phi_0\,(1 - \kappa d'), \tag{3}$$

so κ is the proportional reduction in survival caused by the SUD. The mortality penalty applies here, and here only.

Period 2. Terminal, after which the household dies with certainty. A household without the SUD is unemployed with probability u_0 and one with the SUD with the higher probability u_1; if unemployed it receives income support. The voucher is drawn again, subject to the policy rule. The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and where the SUD is realised. Mortality is a tax on a continuation value, so the SUD can only cost anything if a future remains when it is carried. That does not require a third period. It requires the SUD to be drawn at a *transition* rather than inside the terminal period, and here it is drawn on the way into period 2, with survival over that transition depending on it, so acquiring it puts the whole of period 2 at risk. Realise it inside the last period instead and death costs nothing because nothing follows it, and the mechanism the model rests on falls silent; an earlier version of this note bought its way out of that with a third period, which is the more expensive fix. It is also the less faithful one, since the quantitative model draws every shock after the choices that condition it.

There is accordingly no recovery parameter here. Recovery needs a period to recover into and period 2 is terminal, so ρ has nothing to name. It belongs to the quantitative model, where treatment is a choice and ρ is the parameter its entire effect runs through, and its absence here costs the policy comparison nothing: what separates the two policies is whether the voucher is withdrawn, not whether the household can get clean again.

1.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e=1, \\ \psi \cdot \min\{y, \bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of ψ is what links the SUD to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h \in \{h_0,\; h_S,\; h_L\}, \qquad h_0 < h_S < h_L, \tag{5}$$

homelessness, a small unit and a standard unit. Homelessness is free and delivers almost nothing. Two rented sizes is the fewest that lets a household economise on housing rather than lose it: with one, homelessness comes out at 3.98 per cent against a target of 0.40, at any h_0.

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option j is

$$r_j(\eta) \;=\; p_r\, h_j \big[\kappa_r \eta + (1-\eta)\big] \cdot \mathbf{1}\{j \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and π_V, the share of eligible households actually covered, is the policy dial of Section 1.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of (d', v). **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since cover in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before v is chosen and conditionality cannot screen anyone out on entry. What it does instead is withdraw the voucher from a household that goes on to use, within the same period. So under Treatment First a household that uses pays market rent immediately and again in period 2, and under Housing First it keeps what it drew: the condition is on the choice, never on the arrival.

1.2.4 The household's problem

Let a be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$W_2(a,y,e,\eta,d) = \max_j \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_j + \aleph \Big\}, \qquad c = Ra + y^g(y,e) - r_j(\eta), \tag{8}$$

and write $W_2(a,y,d) \;=\; \mathbb{E}\big[W_2(a,y,e,\eta,d) \;\big|\; y,d\big]$ for its expectation over the period 2 employment and voucher draws. In period 1 the household draws (e,η) and then chooses housing, drug use and saving,

$$V_1(y,e,\eta) = \max_{j,\, v \ge 0,\, a' \ge 0} \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_j + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'} \Big[\phi(d')\, W_2(a', y, d') \Big] \Big\}, \tag{9}$$

with $c = y^g(y,e) - r_j(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2), and $V_1(y) = \mathbb{E}[V_1(y,e,\eta) \mid y]$.

Survival sits *inside* the expectation over d' rather than outside it, and that is not a matter of notation. Survival depends on the realised SUD, so the continuation is $\mathbb{E}_{d'}[\phi(d')W_2]$ and not $\phi\,\mathbb{E}_{d'}W_2$. Written the second way, any household whose continuation value is negative is made better off by a lower survival probability, so it would seek the SUD out for the mortality it brings, and the mechanism the model rests on runs backwards. It is the same failure that makes the level of $\aleph$ a condition to be checked rather than a free normalisation.

A household uses if the value at its best positive v exceeds the value at $v = 0$. Households differ only in y and in the draw (e, η), which is the same lottery for everyone, so within a draw the decision defines a set of income intervals on which use occurs, and participation at a given income is the probability of drawing a state in which the household uses.

1.3 Calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the annual discount factor is 0.995 and the gross interest rate is $1/\beta$, the dispersion of log permanent income is $\sigma = 0.75$ from Heathcote, Perri and Violante (2010), and survival over a period is $\phi_0 = 0.91$, the CDC life table figure from twenty five to fifty five. For the SUD, the proportional fall in survival is $\kappa = 0.43$, derived from Hjorthøj et al.'s (2023) mortality rate ratio rather than assumed: the ratio multiplies the *annual* death hazard, which is then compounded over the thirty years of a period, so a clean household's chance of dying before period 2 is 9 per cent and a household that acquires a SUD faces 48 per cent, and the two unemployment probabilities are measured directly, $u_0 = 0.049$ from the BLS and $u_1 = 0.1625$ from NSDUH 2024 for respondents aged 26 and over.

The annual discount factor is 0.995 rather than the 0.99 the quantitative model uses, and the reason is the same delay that the discussion of θ below returns to. A period here is thirty years, so *every* consequence of using arrives thirty years after the decision, where in the quantitative model consequences arrive annually and compound. At an annual 0.99 the household would weigh that distant death at 0.74 of its value, and the mortality channel would be losing to impatience rather than to anything the model is about. A higher factor is therefore a correction for the coarseness of the horizon rather than a claim about preferences, and 0.995 is as patient as the model can be made: at $\beta = 1$ the poorest household's ex ante value of life turns negative, which reverses the mechanism, and the drug spending share falls further still.

This is the clearest reason the note is only for showing mechanisms, and cannot be used for a policy assessment. A model whose entire lifetime is two dates has to buy the deterrent it needs with a discount factor chosen for the horizon, and the horizon then decides how much a user can spend on drugs, and the two turn out to be the same fact. The quantitative model, where the periods are years and the consequences arrive gradually, is what the paper's numbers come from.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The standard rental unit costs 21.2 per cent of mean income, its rent to income ratio in the ACS, and the small unit is half of it. The voucher parameters are institutional: covered tenants pay 30 per cent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the HUD share of eligible households actually assisted.

Four parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly: with no preference heterogeneity the aggregate is a cliff in b, so a gradient method steps out of the region where participation is interior and walks to a corner. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 5.13 per cent, housing services on the street h_0 to a homelessness rate of 0.399 per cent from the HUD point in time counts, and the SUD hazard λ to the 33.2 per cent of users who acquire a SUD.

The weight on drugs θ is set rather than calibrated, and where it is set is the sharpest thing this model has to say about its own limits. The natural target is the 11 per cent of income that users who are not addicted spend on drugs, which the quantitative model matches. This model cannot reach it while mortality

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a thirty year period	0.99 annually, compounded	0.860
R	Gross interest rate over a thirty year period	Set to $1/\beta$	1.162
σ	Dispersion of log permanent income	Heathcote, Perri and Violante (2010)	0.75
ϕ_0	Survival over a period, no SUD	CDC life tables, ages 25 to 55	0.91
θ	Weight on drug use	Set; see the text	0.20
γ	Weight on housing services	$0.25(1-\theta)$, Consumer Expenditure Survey	0.200
The SUD			
κ	Proportional fall in survival with a SUD	Hjorthøj et al. (2023)	0.43
u_0	Unemployment probability, no SUD	BLS	0.049
u_1	Unemployment probability, with a SUD	NSDUH 2024, ages 26 and over, weighted	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation of the units of v	1.00
h_L	Standard rental unit	ACS 2023, rent over income	0.212
h_S	Small rental unit	Half the standard unit	0.106
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD, share of eligible assisted	0.27

still deters anyone, and the reason is the *delay*.

Everything a household loses by acquiring a SUD arrives one period later, which here is thirty years, and what it loses is a continuation value that the value of a statistical life pins directly: $\beta\,\mathbb{E}W_2/u_c$ is 80 annual incomes by construction, and a SUD destroys the fraction $\kappa\phi_0$ of it. That same product is what makes a poor household use as little as it can while still using, and what puts users at the bottom of the income distribution in the first place, so the drug spending share and the income gradient are not two moments but one. Raising θ buys spending share only until the household stops being deterred, and then it stops buying anything. At $\theta = 0.22$ users are still at a fifth of average income and mortality alone still reverses the gradient; at $\theta = 0.23$ mortality alone no longer reverses it; by $\theta = 0.25$ the spending share is 21 per cent and what little gradient remains is produced by unemployment rather than by mortality. There is no interior point between the two regimes.

So $\theta = 0.20$, with $\gamma = 0.25(1-\theta) = 0.200$, is the largest weight that leaves both results standing, and at it the model produces a spending share of 4.69 per cent against 11. Two periods rather than three raised that from 1.9 per cent, by cutting the delay from two periods to one; the rest of the gap is the delay that remains, and no parameter closes it. That is a statement about what a model with one transition can do, and it is the reason this note is qualitative: it exists to show the direction and relative size of two opposing mechanisms, and the magnitudes belong to the quantitative model, which has sixty annual periods, separate light and heavy experimentation margins, and drug demand from households that already carry a SUD.

Three moments the calibration does not target are worth reporting, since the model is free to miss them. The average user earns 0.19 times average income, which is Section 1.4's result and not a target. The share of the homeless carrying a SUD is 20 per cent, against a fifth to a third in the data depending on whether alcohol is counted, and it is this low only because period 1 carries unemployment as well: every period 1 household is clean, so a model in which only the second period could be homeless would put the share far too high. Against those two, the housing share of income is 16 per cent against a third in the data, which is what a model with two rental sizes and no adjustment margin can do.

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
$\aleph$	Flow value of being alive	Value of a statistical life, annual incomes	80.0	80.0	2.768
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00266
h_0	Housing services when homeless	Homelessness rate	0.399%	0.381%	0.00047
λ	Addiction hazard per unit of use	Share of users who acquire a SUD	33.2%	36.9%	156.3

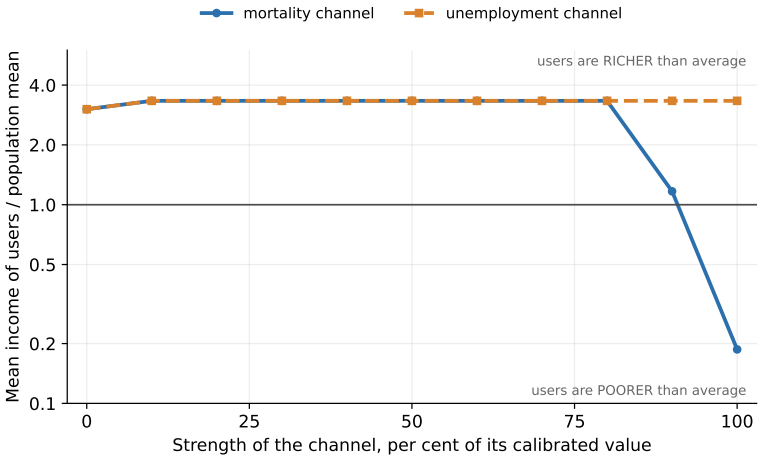


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 per cent throughout.

1.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 3.02 times the population average.

Two features of the environment could reverse that. The *mortality channel* is that the SUD kills you, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to its calibrated value, recalibrating b at every point so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio from 3.02 with nothing switched on to 0.19 at full strength, and it does so as a threshold rather than gradually: at half of its calibrated value the ratio is still 3.33, and the reversal happens over the last tenth or so of the channel. The unemployment channel does not reverse it at all, leaving users at 3.33 times average income even fully switched on. A resource penalty cannot produce the concentration of drug use at the bottom.

Where that threshold sits is not a free feature of the model, and it is the price paid for the drug spending share of Section 1.3. At $\theta = 0.11$ the same two period model reverses the gradient at half the mortality penalty and spends 1.9 per cent of a user's income on drugs; at $\theta = 0.20$ it needs about nine tenths of the penalty and spends 4.69 per cent. A larger appetite for the drug takes more killing to offset, so the two moments trade against each other here in the same way they do everywhere else in this model.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to

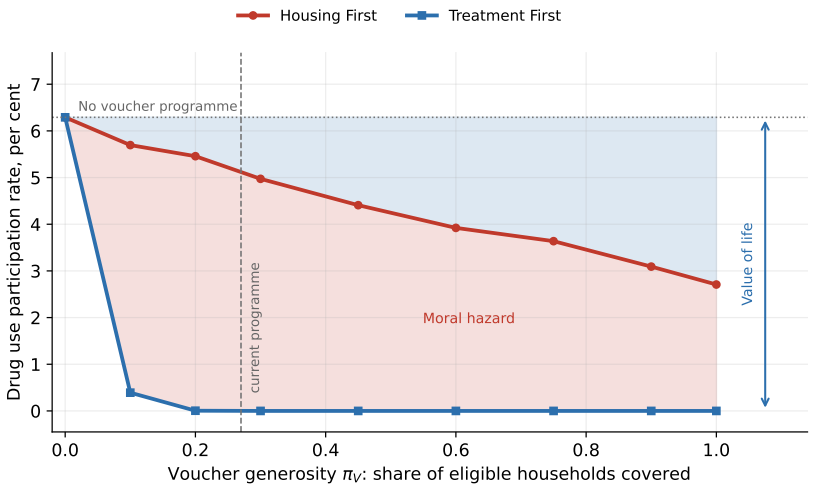


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

1.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 6.29 per cent of households use. Universal Treatment First takes that to 0.00 per cent, a value of life force of -6.29 percentage points; universal Housing First leaves it at 2.71 per cent, so the moral hazard force is +2.71 points and the net effect of Housing First is -3.58. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. On homelessness the two policies end in the same place, both driving it from 0.77 per cent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing at risk. The larger difference is on the other margin, where at universal cover Housing First leaves 0.372 per cent of the population carrying the SUD and Treatment First leaves 0.000 per cent. What the moral hazard buys is extra SUD cases among households that are housed either way.

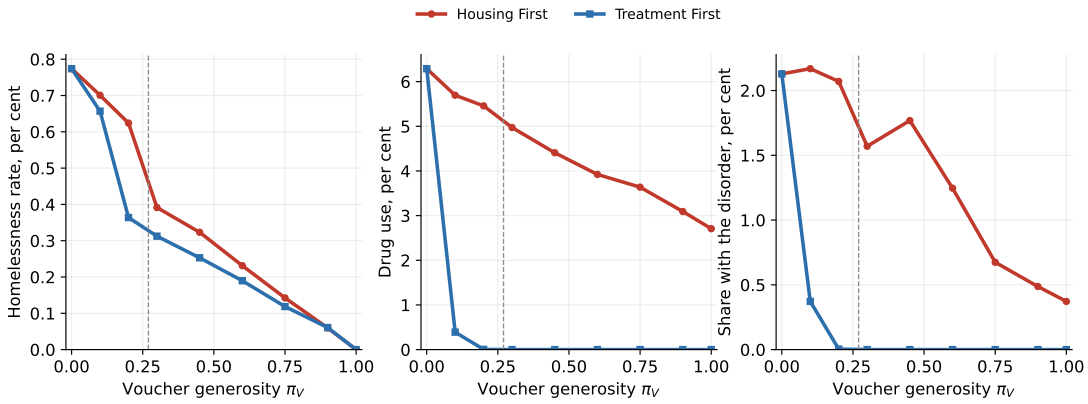


Figure 3: Homelessness, drug use and the SUD under the two policies

1.6 Welfare

1.6.1 Without externalities, Housing First always wins

Sections 1.4 and 1.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off:

$$V_1(y;\ \text{HF},\ \pi_V)\ \geq\ V_1(y;\ \text{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[1]

It follows that the moral hazard is not a distortion. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +2.71 points of Section 1.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 4.1 per cent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.29 per cent of consumption.

A fixed budget does not rescue conditionality either. Treatment First is actually slightly *more* expensive at universal cover, 0.1717 against 0.1716, so matching on cost rather than on generosity changes nothing. The reason is that survival into period 2 falls from 0.91 to 0.52 with a SUD, so the households the moral hazard adds to the rolls die sooner and draw voucher payments in fewer states. So a case for conditionality has to come from an externality.

1.6.2 The externality on children reverses it

The externality I model is one that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1+v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. Letting it inherit a greater susceptibility, a higher λ, fails the other way, since a child facing a riskier drug optimally uses *less* and is worse off ex ante, which is clearly counterintuitive.

Income works on both counts, and it works through the channels explored in Section 1.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born

[1] The same comparison in the quantitative model gives the same answer.

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	2.71%	0.00%
Share with a SUD	0.372%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.1716	0.1717
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		4.1%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.29%
Consumption equivalent loss, whole population		0.012%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household from period 2 on, discounted, per unit of average income.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's SUD rate	Ratio	Loss to the child
0.00	2.23%	1.00	0.0%
0.20	4.02%	1.80	33.3%
0.30 (calibrated)	5.43%	2.43	59.2%
0.40	7.39%	3.31	96.2%
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.012%
Disorder cases prevented			0.372 pp
Damage to the child of one such case			47%
Externality gain from conditionality			0.175%
Gain over loss			14.5

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

poorer is mechanically more likely to use, which is aligned with the data. The income gradient does the whole job, so the model transmits a SUD across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1.

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the SUD 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty δ_h is set directly from Chetty, Hendren and Katz (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 per cent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.012 per cent of consumption, small because only 4.1 per cent of households are affected. Against that, conditionality prevents 0.372 percentage points of SUD cases in the second row, and each of those cases costs one child 47.1 per cent of consumption, in the third. The externality gain is 14.5 times the private loss, and stays above it throughout the range of δ_d in Panel A. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

Treatment First drives drug use to *exactly* zero here, and that inflates the margin. Nobody in this model faces the sobriety condition and fails to meet it, and the households who cannot comply are the entire empirical case for Housing First.

The point of this section is only that an externality can reverse a ranking that is otherwise airtight, and one externality is enough to show it. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model is for.

2 The quantitative model

This section states the model the paper's numbers come from, completely. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below exactly as Sections 1.2 to 1.6.2 give them. Everything else is written out, because it is what turns a directional answer into a quantitative one.

2.1 Demographics, the state and the choice set

Households enter at age 26, live at most to 85, so $j = 1, \ldots, J = 60$ annual periods, and retire at $j_R = 41$. Survival from j to $j+1$ is $\phi_j(h', d, m)$ in (27): it depends on age, on the housing *chosen* this period, on the SUD and on mental illness. Newborns replace the dead. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

assets $a \geq 0$ on a 50 point grid dense near zero, permanent productivity y on 7 points, employment e, the SUD d, serious mental illness m and the voucher η, which is 8,400 points per age and 504,000 in total. Assets are non negative throughout: with homeownership out of the model there are no mortgages, so the grid has no negative side.

Two economies in that list are choices rather than oversights. There are two employment statuses, employed and unemployed. Unemployment insurance is paid for the first year of a spell and nothing after it, so a household in its first year of unemployment and one in its third have different income and different continuation values and cannot be merged; e therefore takes three values, the third being an unemployment duration marker rather than a third status. There is no probability of receiving insurance: it used to be drawn on entry into a spell, and that parameter is gone, which removes a parameter without removing a state. And the housing unit a household arrives in is not a state: everything that conditions on housing conditions on the unit chosen, and renters face no adjustment cost, so carrying it would replicate the whole problem four times over for nothing. It would have to be reinstated if moving costs were added.

Each period the household chooses

$$\left(n,\, h',\, v,\, \tau,\, a'\right) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

hours n, housing h' where h_1 is homelessness, drug use v, treatment take up τ, available only when $d = 1$, and next period assets a'. Every shock is drawn *after* all of that, which is what makes those variables state variables rather than choices, and it is why the transitions in Section 2.10 condition on the chosen h', v and τ: the choice comes first, the draw second.

2.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\left[\, c^{\alpha}\,(h')^{\beta}\,(v + b_{dmh'})^{\gamma}\,\right]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\bar{\tau}^{u}, \tag{14}$$

with $\alpha + \beta + \gamma = 1$. Equation (1) is the logarithmic case of the bundle with hours fixed, so γ and b do what θ and b do in Section 1.2: b is the Stone-Geary endowment that keeps the marginal utility of the first unit finite and therefore decides whether anyone uses at all.

Three things are new. Leisure makes hours a margin. Treatment costs utility rather than money, which is what the evidence supports, since the three largest barriers reported in the NSDUH are beliefs and preferences and not prices. And the Stone-Geary shifter itself varies,

$$b_{dmh'} = b_d \cdot \left(\rho_m^b\right)^{\mathbf{1}\{m=1\}} \cdot \left(\rho_H^b\right)^{\mathbf{1}\{h'=h_1\}}, \qquad \rho_m^b, \rho_H^b \in (0,1], \tag{15}$$

so a SUD, mental illness and homelessness each raise the marginal utility of the first unit of the drug. The Cobb-Douglas weights themselves are held fixed across the SUD: only the shifter moves. That is a response to Referee 2 (C5), Referee 3 (M2a) and the AE (A8), who all objected that a taste weight rising with d makes reducing the SUD and expanding housing complements even absent any incentive effect. With the weights invariant, acquiring a SUD relaxes no constraint, so every private cost of it has to come from mortality, employment and the budget. Varying b instead is the AE's own suggested alternative, A8.

Three separate things happen in equation (15). First, b_0 against b_1: a single b cannot match participation and the expenditure share at once, because a b large enough to deter clean households pushes the interior choice below zero for households that already carry a SUD and sends their expenditure to the corner, so b_0 pins participation and b_1 keeps the choice interior for θ_v to pin. Second, $\rho_m^b < 1$ is self medication, the channel Referee 1 asks about in comment 1(a). Third, and most important, $\rho_H^b < 1$ is the housing access channel: it is the *only* parameter in the model that governs the elasticity of drug use with respect to housing access, which Referee 2 (C1) calls the crux on which all the welfare results rest and which Referee 3 and the AE (A13) name independently. Without it that moment is not merely untargeted, it is untargetable, since housing a household only raises its resources and drugs are a normal good, so the model would predict that housing *raises* use, the opposite sign to Milby et al. (2005).

2.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq j_R \quad \text{(social security)} \\ n\,y\,\varepsilon_j\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j < j_R,\ e=1 \quad \text{(employed)} \\ \min\{\varrho_{UI}\,y\,\varepsilon_j,\ \bar{u}\,\bar{Y}\} & j < j_R,\ e=2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\,\bar{y}_{DI}\,\bar{Y} & j < j_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{16}$$

with ε_j the deterministic age profile of efficiency, ρ_d and ρ_m productivity penalties applied *conditional on working*, ϱ_{UI} the replacement rate paid on the household's own productivity, up to a cap $\bar{u}$ on the benefit as a share of average income, as in equation (4), ζ_{DI} the share of the non employed with mental illness who receive disability, and $\bar{Y}$ average income. Retirement income replaces 40 per cent of the productivity state with no health penalty, since social security is not means tested on health.

The conditioning of the penalties on *working* is what lets an employed household be poor enough to choose homelessness. An unemployment effect alone cannot do that, and the data require it: the homeless are roughly half in work, not none. Section 1.2.3's environment, where only unemployment lowers income, is exactly the version that cannot.

Labour income is taxed with the Heathcote, Storesletten and Violante function and capital income at a flat rate,

$$y^n = \tau_\ell\,(y^g)^{\xi} \;+\; (1-\tau_k)\,r\,a, \tag{17}$$

where ξ governs progressivity and τ_ℓ the level, and τ_ℓ is what clears the government budget. Two transfers follow, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\,\bar{Y}$, and the voucher covers rent above a tenant contribution of κ_r of gross earnings,

$$F = \max\left\{0,\ \underline{c}\,\bar{Y} - (y^g + \max\{0, ra\})\right\}, \qquad Vch = \mathbf{1}\{\eta=1\}\,\mathbf{1}\{h' \neq h_1\}\,\max\left\{0,\ p_h h' - \kappa_r\,y^g\right\}. \tag{18}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

2.4 The budget constraint and the hours choice

$$c + p_v v + a' \;=\; y^n + a \;-\; p_h h' \,\mathbf{1}\{h' \neq h_1\} \;+\; Vch + F, \qquad c > 0,\; v \geq 0,\; a' \geq 0. \tag{19}$$

Note what is absent: treatment, which costs utility, and bequests, which go to the government. Given a' and h', write R for the resources left over. Cobb-Douglas preferences over $(c, v + b)$ split them in closed form,

$$c = \frac{\alpha}{\alpha+\gamma}\big(R + p_v b_{dmh'}\big), \qquad v = \max\Big\{0,\; \frac{\gamma}{(\alpha+\gamma)p_v}\big(R + p_v b_{dmh'}\big) - b_{dmh'}\Big\}, \tag{20}$$

so the inner allocation never needs a numerical search, and the corner in v is where the household abstains because the Stone-Geary endowment already exceeds its desired intake. That is how a large b deters use, and it is the same mechanism as the threshold in Section 1.2.4.

Hours solve

$$u_c(c)\,\frac{\partial R}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}. \tag{21}$$

The clawbacks in (18), at a rate of one for food stamps and κ_r for the voucher, make $\partial R/\partial n$ jump *up* at the hours where each transfer is exhausted, because the clawback stops. The residual is therefore not monotone in n and (21) can have several roots, so it is solved by comparing the candidate roots rather than by a first order condition alone. This is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

2.5 Housing, and the homelessness loop

Housing is a four point grid. h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point in time definition and excludes doubling up; it carries a small service flow and no rent. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{22}$$

so two parameters describe the grid. Rent p_h is normalised to one and the *grid* is the calibrated object, since rescaling the grid by λ and the price by $1/\lambda$ leaves every budget constraint untouched and identifying the level would run through a weak channel.

Homelessness then enters four separate equations: survival (27), the employment transitions (29), the recovery hazard (24) and the Stone-Geary shifter (15). It raises mortality, worsens both employment margins, lowers the chance of recovering, and raises the marginal utility of the drug. Those four together are what make homelessness and the SUD mutually reinforcing rather than merely correlated, and it is that loop, absent from Section 1.2, that a housing programme acts on.

2.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with v chosen continuously by (20). A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{23}$$

a single hazard rising in the quantity used, with no light and heavy split and no threshold, so as in Section 1.2 the intensive and extensive margins are not separate decisions. In the baseline a clean household may initiate only up to an age j_E, which Referee 1's smaller comment 1 notes is in tension with the U shaped age profile of use. It is a switch rather than a hard coded parameter, so the model runs both ways and the age profile is reported both ways. A household with the SUD recovers with probability

$$\pi_r(h', \tau, m) = \min\Big\{1,\; \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\Big\}, \tag{24}$$

with $\mu_H < 1$, recovery is harder on the street, $\mu_m < 1$, harder with mental illness, and $g > 1$ the treatment effect.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). This matters more than it looks: the policy under evaluation is called Treatment First, and g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline. Section 1.5, where the only margin is whether to use, has no counterpart for it.

2.7 Serious mental illness

m is an exogenous two state Markov chain representing *serious* mental illness, about 5.5 per cent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population. The onset rate is derived from the target prevalence p_m and the recovery rate λ_m,

$$\text{onset}_m = \lambda_m \frac{p_m}{1-p_m}, \qquad \Pr(m'=1 \mid m=0,d) = \text{onset}_m \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m'=0 \mid m=1) = \lambda_m, \tag{25}$$

where the onset rate is not a scalar. A flat onset against a flat recovery drives prevalence towards its steady state from below, so a scalar produced a prevalence profile that *rises* with age, where NSDUH has it falling from 11.0 per cent at ages 26 to 29 to 1.4 per cent at 65 and over, a factor of eight. Onset is therefore backed out of the observed prevalence profile and the recovery hazard,

$$\lambda(a) = \frac{p(a{+}1) - p(a)(1-\mu)}{1-p(a)}, \tag{26}$$

with prevalence fitted log linearly, $R^2 = 0.977$, rather than interpolated piecewise: differencing a piecewise linear curve turns every kink into a spike, and an early attempt gave onset of 1.00 at 26, 0.01 at 32 and exactly zero at 56 and 68, which are artifacts rather than incidence that vanishes and returns. As with the mortality multipliers, the data supply the shape and the level stays calibrated to the prevalence target, and newborns enter at the observed prevalence at 26 rather than at the ergodic distribution of the age one transition matrix, which would start them at 4.7 per cent against 11 to 13 observed. So calibrating persistence never disturbs prevalence, and $\iota > 1$ makes a SUD raise the hazard of mental illness. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter, and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74.

Mental illness then affects six objects: productivity conditional on working through ρ_m in (16), both employment margins through κ_{uem}, mortality through $\kappa_{\phi m}$, the recovery hazard through μ_m, the Stone-Geary shifter through ρ^b_m in (15), and eligibility for disability income through ζ_{DI}.

The mental health chain does *not* depend on housing, and that is a result rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with mental illness across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

2.8 What the SUD and homelessness cost

Survival is

$$\phi_j(h',d,m) = 1 - \min\left\{1,\ (1-\bar{\phi}_j)\,\kappa_{\phi H}^{\mathbf{1}\{h'=h_1\}}\,\kappa_{\phi d}^{\mathbf{1}\{d=1\}}\,\kappa_{\phi m}^{\mathbf{1}\{m=1\}}\right\}, \tag{27}$$

with $\bar{\phi}_j$ the baseline age profile and the multipliers themselves varying with age, rather than the single κ of (3). Because $\aleph > 0$ a higher death hazard is a real cost, and it is larger for households with more life remaining and more to consume, which is precisely the sorting Section 1.4 isolates.

The hazards vary with age, and three of them had to. Each was previously a constant assigned to hit an average, so each matched a level while getting the life cycle profile wrong, in every case in the direction that weakens the mechanism at the ages where it does its work.

$\bar{\phi}_j$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$, rather than averaged equally: by the mid eighties the two survivor populations differ enough that equal weights are simply wrong. The smooth functional form it replaces understated mortality by nearly half at the top, $q = 0.045$ at 85 against an actual 0.081 (Referee 3, M1b).

The homelessness and SUD multipliers cannot be set separately, because Baggett et al.'s homelessness ratio is measured on a population that mostly also carries a SUD. Netting that out gives

$$\kappa_{\phi d}(a) = 1 + (\kappa_{\phi d} - 1)\,\sigma_d(a), \qquad \kappa_{\phi H}(a) = \max\left\{1,\ \frac{RR(a)}{(1-s) + s\,\kappa_{\phi d}(a)}\right\}, \tag{28}$$

with σ_d the SUD profile from CDC drug poisonings normalised to one at its peak, RR Baggett's all cause ratio, and $s = 0.217$ the model's own SUD share of the homeless. Because $\kappa_{\phi d}$ falls with age so does the correction, so at 45 to 64 more of the observed ratio is genuine homelessness effect and $\kappa_{\phi H}$ at 56 comes out at 3.14 rather than the 1.97 a flat calculation gives.

The *level* of $\kappa_{\phi d}$ stays free on purpose. It is the second stage's only instrument: it is bracketed inside the Degenhardt SMR range to hit a drug price participation elasticity of -0.63, and that elasticity is what disciplines how much drug use responds to its costs, which is the object the whole policy experiment turns on. Pinning the level from mortality data would take the instrument away and leave the elasticity undisciplined, so the data supply the profile and the elasticity supplies the level. The system is then over identified, which is the point: the age weighted SMR the model produces is reported against the published ratios, and a disagreement is a finding rather than something a dual purpose parameter can hide.

$\kappa_{\phi m}$ is the one that stays flat, and that is a result rather than a default: Olfson et al. (2015) give all cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising. The employment hazards stay constant too; job finding does fall with age in the data, but it shapes the income profile rather than any of the paper's mechanisms.

Job finding and separation are

$$\pi^{ue} = \min\left\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\right\}, \qquad \pi^{eu} = \min\left\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\right\}, \tag{29}$$

so each multiplier below one makes work simultaneously harder to find and easier to lose. In the baseline there is no direct wage penalty conditional on working, $\rho_d = 1$: the robust findings are on the employment margin, roughly half again as much unscheduled absence and 44 per cent higher turnover, while the wage penalty conditional on working is contested. A version calibrated to the income ratio of those with the SUD is reported as an alternative.

2.9 The voucher, and the two policies

A household without a voucher draws one with probability π_V if it is income eligible, and a household with one keeps it while eligible, so unlike Section 1.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + \max\{0, ra\} \ < \ \kappa_V \cdot \text{median income}. \tag{30}$$

Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that used and with probability p_{fp} by one that did not,

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{HF} \times \begin{cases} 1 - p_{\text{det}} & \text{if } v > 0, \\ 1 - p_{\text{fp}} & \text{if } v = 0. \end{cases} \tag{31}$$

This is the one place where the quantitative model changes the *logic* of Section 1.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers are illustration.

2.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle. It carries the cross sectional dispersion of permanent income and the intergenerational link, and it is a type rather than a process, so $y' = y$. This is a departure from the submitted paper, where y followed a persistent autoregression, and it is made for two reasons: it is what keeps the child's penalty (37) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, and unemployment is the income risk that links the SUD to housing, which is retained in full. The stochastic version is the robustness direction.

Employment e. A household that separates receives insurance with certainty for one year, so with π^{ue} and π^{eu} from (29),

$$e = 1 \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e \in \{2, 3\} \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}, \end{cases} \tag{32}$$

which is where insurance lasting exactly one year comes from: state 2 is entered only from employment and is never entered from itself, so the second year of a spell is state 3 and pays nothing.

The SUD d. By (23) if $d = 0$ and (24) if $d = 1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (25), conditioning on d.

The voucher η. By (30) and (31), conditioning on next period's eligibility and, under Treatment First, on the chosen v.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (33) is taken against.

2.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with V_{J+1} given below,

$$\begin{aligned} V_j(a, y, e, d, m, \eta) = \max_{n,\, h',\, v,\, \tau,\, a'} \Bigg\{ & \frac{\big[c^{\alpha}(h')^{\beta}(v + b_{dmh'})^{\gamma}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} \\ & + \beta\,\phi_j(h', d, m)\int V_{j+1}(s')\,\Gamma\big(\mathrm{d}s' \mid s, n, h', v, \tau\big) \\ & + \beta\,\big[1 - \phi_j(h', d, m)\big]\,\omega\,\mathbb{E}\,V_1\big(0,\, y_{\text{child}},\, 1, 0, 0, 0\big) \Bigg\} \end{aligned} \tag{33}$$

subject to the budget (19), the income definitions (16) to (18), the choice set (13), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq j_R$ or $e \neq 1$, $a' \geq 0$, and $V_{J+1} = 0$ with $\phi_J = 0$, so that at the last age the child is the whole continuation.

The altruism is carried at *every* age and not only at the end. A household that does not survive from j to $j + 1$ leaves a child, so the continuation is the survival weighted value of its own next period plus the death weighted value of the child's life, each discounted, with y_{child} drawn from (37). That matters because ϕ_j itself depends on h' and, through d, on v: a household that uses raises the chance it is the child's term rather than its own that is realised, so the child enters the drug and housing decision through the weight on the two terms as well as through y_{child}. It also means the value of dying is $\omega\,\mathbb{E}V_1$ rather than zero, which is a condition to check rather than a normalisation: at a large enough ω a household whose own continuation

is worth less than its child's would want the extra mortality a SUD brings, which is the failure Section 1.2.4 guards against in the stylized model.

At $\omega = 0$ death is worth zero and the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share ω of it, and only $1 - \omega$ remains external. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on v and h', that last term is what puts the child inside the parent's drug and housing decision.

The problem is solved by exact backward induction. Within a housing choice the split of c and v is closed form by (20) and hours come from (21), so the only search is over a', and it is done on the full grid rather than by a coarse then refine heuristic, because the objective is *not* single peaked in a': eligibility (30) is a step function of income, which includes ra; the food stamp floor kinks; and housing is discrete.

2.12 The government

The government pays social security, insurance, disability, food stamps and vouchers, and bears a fiscal cost χ_H per homeless household and χ_D per household with the SUD, including the cost of the drug tests that make detection possible. It finances this with the two taxes in (17) and the estates of the deceased, and the level parameter τ_ℓ clears the budget,

$$\underbrace{\sum y^g - \tau_\ell \sum (y^g)^\xi + \tau_k \sum \max\{0, ra\} + \text{estates}}_{\text{revenue}} = \underbrace{SS + UI + DI + F + Vch + \chi_H H + \chi_D D + G}_{\text{outlay}}. \quad (34)$$

Estates accrue to the government rather than being redistributed as a lump sum, because a uniform transfer to the poor has welfare consequences of its own that would contaminate every policy comparison in the paper, and because it removes an equilibrium fixed point.

Equation (34) is the single most important thing the stylized model does not have. There, the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, τ_ℓ rises to pay for it, and the higher tax distorts (21) and saving for everyone, including the households the programme never reaches. That distortion has no counterpart in Section 1.6, it runs against Housing First, and it is the main reason the dominance result (10) does not survive into the quantitative model intact.

2.13 The drug market

Drug production has cost $C(V) = \frac{\psi_1}{\psi_2} V^{\psi_2}$, so supply satisfies

$$p_v = \psi_1 V^{\psi_2 - 1}, \qquad V = \int v(s)\, \mathrm{d}\Phi(s), \quad (35)$$

with price elasticity of supply $1/(\psi_2 - 1)$ and ψ_1 normalised so that $p_v = 1$ at the benchmark quantity. Endogenising the price does not suppress the normalisation of Section 1.2.3, it moves it to ψ_1.

The baseline elasticity is 2, so $\psi_2 = 1.5$. There is no direct estimate for an illicit drug, so it is taken from the closest legal markets, where the estimates run from about 0.7 to about 4: the Washington State marijuana reform implies roughly 0.79, and the cigarette and gasoline literatures put it nearer 3.6. Two is the middle of that range and the range itself is reported. The perfectly elastic case is its upper end rather than the baseline, and it has a natural experiment behind it. In 2010 the reformulation of OxyContin pushed a very large mass of demand into heroin, enough to quadruple heroin death rates, and purity adjusted heroin prices did not rise; they fell slightly. A demand shift of that size leaving price flat is evidence that the curve is close to horizontal, which is why the top of the range deserves as much weight as the bottom. What a finite elasticity buys is that a policy which lowers aggregate drug use also lowers its price, and the cheaper drug offsets part of the fall, so every effect on drug use below is smaller than it would be at a fixed price.

2.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi_j \int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\, \mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{36}$$

where ϕ_j is evaluated at the household's own $(h'_j(s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, employed, clean, without mental illness and without a voucher, whose productivity is drawn from the unconditional distribution Ω_y scaled by the penalty,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{37}$$

This is equation (11) made operative, and two implementation points matter. The penalty shifts the newborn's draw *probabilities* over the existing productivity nodes rather than creating penalised types, which would multiply the productivity space by four for no gain. And the parent's status is read at death, which understates exposure, since a parent who acquired the SUD at forty and recovered at sixty leaves no mark; the resulting externality is therefore a lower bound.

2.15 Stationary equilibrium

A stationary equilibrium is a set of value functions $\{V_j\}$, policy functions $\{n_j, h'_j, v_j, \tau_j, a'_j\}$, a distribution Φ, and prices and aggregates $\big(\tau_\ell,\, p_v,\, \bar{Y},\, \text{median}\big)$ such that

1. the policy functions solve (33) given the terminal condition (**??**), and $\{V_j\}$ are the associated values;
2. Φ is invariant under the law of motion (36) together with the newborn rule (37), and the mass of newborns equals the mass of the deceased;
3. the government budget (34) holds at τ_ℓ;
4. the drug market clears at p_v, so (35) holds with $V = \int v(s)\,\mathrm{d}\Phi$;
5. $\bar{Y}$ and the median of gross income are the aggregates implied by Φ.

Two of those are genuine fixed points rather than accounting. The median enters eligibility (30), and in a skewed income distribution it sits well below the mean, so using the mean would materially overstate the eligible population. And τ_ℓ is what makes the policy comparison a general equilibrium object at all: the two policies do not cost the same once take up and detection are imperfect, so they do not face the same tax, and the household problem is different under each for reasons that have nothing to do with the voucher rule itself.

2.16 What is left out, and what it would do

The model has no disamenity from aggregate homelessness: households do not care how many other households are on the street. That was in an earlier version as a utility cost $\kappa_x(H + \omega_x H_D)$ and it is now out of the model rather than set to zero, because the evidence says it cannot change an answer. Loubière et al. (2020) ask adults in eight European countries what they would pay in annual taxes to end homelessness through Housing First: the mean over all respondents is €23 a year and €57 among the 42 per cent willing to pay anything, which is 0.08 and 0.20 per cent of average annual earnings. A term that small cannot move a ranking, and carrying it would put the homelessness rate inside the utility function and so add a fixed point to the equilibrium for nothing. Leaving it out is conservative in a known direction: it can only understate the case for whichever policy houses more people, so the reported results are a lower bound on the case for housing.

3 Calibrating the quantitative model

Two stages, and they are different exercises. The first matches a cross section of the stationary equilibrium, nineteen parameters against twenty one moments. The second matches short run *responses* measured in

experiments, which cannot be read off a comparison of steady states. Only what is new or what Section 1.3 does not already illustrate is described here.

The same cliffs, and the same answer. Three moments are step like and are bisected rather than stepped, for the reason Section 1.3 gives: participation is a near threshold function of b, homelessness moves from 0.12 to 3.29 per cent for a 35 per cent change in h_{HL}, and wealth over output is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves. The parameter to moment map is close to triangular and the sign of $\partial M/\partial\theta$ is declared for every pair, so a mistaken sign shows up as a moment walking away from its target rather than as a silent failure.

Anchors. The discount factor targets aggregate wealth at three times output, with output measured as all factor income and transfers excluded, which is the standard US private capital to output ratio rather than the SCF's net worth to household income ratio of about 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold. As in Section 1.3, $\aleph$ is set to a value of a statistical life rather than to a behavioural moment, because a behavioural target pins $\aleph$'s effect on *who* uses without pinning its level, and at the wrong level households prefer death and the mortality cost of drug use becomes a benefit.

Three targets that were wrong and are now right. Substance use disorder among the homeless is 21.7 per cent, the *drug* use disorder figure from Gutwinski et al.'s pooling of thirty nine studies, not the 36.7 per cent that is alcohol. The income of drug users is split: 95.1 per cent of average for users and 68.7 per cent for a moderate or severe SUD, ages 26 and over, with the SUD ratio targeted and the use ratio kept for validation, which makes the economics explicit, since use is close to income neutral and it is progression to the SUD that concentrates at the bottom. The housing burden is targeted over all households rather than over renters only.

Second stage: responses, not steady states. These are partial equilibrium cohort experiments. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at benchmark; the household problem is solved again, because each arm changes the rules a forward looking household faces. Milby et al. (2005) randomised 196 cocaine dependent homeless adults into three arms on identical day treatment, and the targeted moment is the ratio of the unconditional voucher arm to no assistance, $6.60/4.48 = 1.473$ weeks abstinent. The conditional against unconditional comparison is deliberately *not* targeted, because the source reports it as insignificant in the raw randomised comparison; calibrating to it would repeat the error this revision is correcting elsewhere. The drug price participation elasticity, -0.63, is the second stage's other target.

What the two new parameters are matched to. The weight on the child ω and the transmission penalty δ_d are fitted *together*, because each moves the other's moment. Altruistic parents use less, so the share of children born to a parent with the SUD falls, so the transmission ratio at a given δ_d falls with ω.

Parameter	Moment	Source
ω	Drug use falls 15.5 per cent with a child present	NSDUH 2024, ages 26+, parents reweighted to the non parents' joint age, income, sex and marital distribution
δ_d	Child's SUD rate relative to a child of a clean parent	The epidemiological range of two to four
δ_h	Set, not calibrated	Chetty, Hendren and Katz (2016)

Since every household in the quantitative model has a child, there is no parent and non parent comparison to run inside it. The condition is the counterfactual instead: ω is the entire reason a household uses less than it otherwise would, so switching altruism off at the calibrated parameters must raise drug use by 18.4 per cent. Run through the stylized model that condition gives $\omega = 0.014$, which is worth stating because it is small: the ranking in Section 1.6.2 reverses back to Housing First only near $\omega = 0.43$, so the calibrated weight sits a factor of thirty inside the region where the externality still decides the comparison, and it is

consistent with Altonji, Hayashi and Kotlikoff (1997), who reject altruism operating at the margin at all. The measured gap is an upper bound, since users also lose custody of children, so the ranking is reported across zero to that value.

Tables 5 and 6 list every parameter of the model: those that are assigned, whether by normalisation, from the literature, directly from data, in equilibrium or as a reported sensitivity, and those that are calibrated inside the model, each against the moment it is assigned to. The full derivation behind each value, which used to sit inside the tables, is in `writing/old/`; what is printed here is the parameter, what it means, where the number comes from and the number itself.

Table 5: Assigned parameters of the quantitative model

Parameter	Description	Source or rationale	Value
Preferences and demographics			
σ	Risk aversion	Attanasio and Weber (2010); the standard life cycle value	2.00
σ_ℓ	Curvature on leisure	Frisch elasticity of 0.5, Chetty (2012)	3.00
ε_j	Deterministic age profile of efficiency	Quadratic in age, fitted to the PSID and CPS hump	profile
$\bar{\phi}_j$	Baseline survival by age	SSA period life table, ages 26 to 85, the sexes combined by survivor weight	profile
ρ_y, σ_y	Persistence and dispersion of log productivity	Storesletten, Telmer and Yaron (2004); the stochastic process is the robustness direction, the baseline holds y fixed	0.90, 0.20
j_E	Maximum experimentation age	Reported with and without it	55
Prices, taxes and the government			
r	Risk free rate	Kaplan, Mitman and Violante (2020)	0.03
ξ	Tax progressivity	Heathcote, Storesletten and Violante (2017)	0.85
τ_k	Capital income tax	US long run rate on capital gains and qualified dividends	0.15
τ_ℓ	Tax level	Clears the government budget	in eqm.
G	Other public spending	Residual at the benchmark, then held fixed so that policy is financed by τ_ℓ	in eqm.
χ_H	Fiscal cost per homeless household	Abramson and Van Nieuwerburgh (2024), the literally homeless figure	0.4286
χ_D	Fiscal cost per household with a SUD	CEA (2017), about \$30,000 per person year	0.212
Income support and the voucher			
ϱ_{UI}	Unemployment insurance replacement rate	DOL statutory replacement rates	0.50
$\bar{u}$	Cap on the insurance benefit, share of average income	DOL state benefit maxima	1.00
$\bar{y}_{DI}$	Disability benefit level, share of average income	SSA, average SSDI benefit against average earnings	0.25
κ_V	Voucher eligibility threshold, share of median income	HUD rule	0.50
κ_r	Tenant contribution rate	HUD rule	0.30
p_{test}	Probability a voucher holder is tested	Benchmark value, swept downward	1.00
c_{test}	Cost per voucher holder tested	A urine panel at about \$40, quarterly	0.0023
p_{det}	Detection probability given a test	Benchmark value, swept downward	1.00
p_{fp}	False positive rate under Treatment First	Benchmark value, swept upward	0.00
Employment, the SUD and mental illness			
$\bar{\pi}^{ue}$	Job finding rate, annual	Krueger, Cramer and Cho (2014)	0.55
κ_{ueH}	Employment multiplier, homeless	Desmond and Gershenson (2016)	0.80
$\kappa_{\phi H}$	Mortality multiplier, homeless	Baggett et al. (2013), varying with age	4.41 to 1.09
$\kappa_{\phi m}$	Mortality multiplier, mental illness	Olfson et al. (2015); flat in age	2.50
ρ_d	Productivity penalty of a SUD, conditional on working	No direct wage effect in the baseline: the employment margin carries it	1.00
$\bar{\pi}_r$	Baseline recovery hazard when housed, annual	Vaillant (1995); bracketed by Fleury et al. (2016)	0.025
μ_m	Recovery multiplier, mental illness	Dual diagnosis relapse of 39.8 against 26.4 per cent	0.82
g	Treatment effect on recovery	Odds ratio of 1.22 for contingency management	1.35
λ_m	Recovery rate from mental illness, annual	İmrohoroğlu and Zhao; about half improve within ten years	0.067
ι	SUD multiplier on the onset of mental illness	Danish registers, hazard ratios of 1.34 to 2.74	2.00
The drug market			
p_v	Price of drugs	Clears the drug market, and is one at the benchmark	1.00
ψ_1	Drug supply shifter	Normalisation, so that $p_v = 1$ at the benchmark quantity	norm.
ψ_2	Drug supply curvature	Elasticity $1/(\psi_2 - 1) = 2$; Hansen, Miller and Weber (2018) give 0.79 and the cigarette and gasoline literatures about 3.6	1.50

Table 6: Internally calibrated parameters of the quantitative model

Parameter	Description	Target moment	Source	Data	Model	Value
ω_ℓ	Weight on leisure	Average hours worked	Guvenen et al. (2023)	40%	—	—
θ_h	Cobb-Douglas weight on housing	Aggregate housing cost over income	SCF 2022	21.2%	—	—
θ_v	Cobb-Douglas weight on drugs	Drug spending over income given a SUD	NSDUH and ADAM	44%	—	—
b_0	Stone-Geary shifter, clean	Drug participation rate	NSDUH 2024, ages 26+	5.13%	—	—
b_1	Stone-Geary shifter, with a SUD	Abstinence among homeless adults with a SUD	Milby et al. (2005), no housing arm	18.7%	—	—
ρ_m^b	Shifter multiplier, mental illness	SUD rate given mental illness	NSDUH 2023	9.75%	—	—
ρ_H^b	Shifter multiplier, homelessness	Housed against unhoused abstinence	Milby et al. (2005)	1.473	—	—
$\aleph$	Flow value of being alive	Model implied value of a statistical life	Viscusi and Aldy (2003)	\$7.5m	—	—
$\bar{\tau}^u$	Utility cost of treatment	Treatment take up given a SUD	NSDUH 2023	24%	—	—
β	Discount factor	Wealth over output	BEA fixed assets against GDP	3.0	—	—
h_{HL}	Service flow of homelessness	Homelessness rate	HUD point in time 2023, households	0.399%	—	—
$\underline{h}$	Smallest rental size	Share with housing costs above half of income	SCF 2022	17.49%	—	—
ς	Geometric spread of the rental sizes	Share with housing costs above 30 per cent of income	SCF 2022	40.62%	—	—
$\underline{c}$	Food stamp consumption floor	Food stamps over labour income	USDA, SNAP outlays	0.42%	—	—
π_V	Voucher lottery probability	Recipients over the income eligible	HUD Picture of Subsidized Households	26.6%	—	—
$\bar{\pi}^{eu}$	Job separation rate	Unemployment rate	BLS, ages 25+	4.9%	—	—
κ_{ued}	Employment multiplier, SUD	Unemployment rate given a SUD	NSDUH 2024, ages 26+	16.25%	—	—
κ_{uem}	Employment multiplier, mental illness	Unemployment rate given mental illness	NSDUH, by Bayes	7.1%	—	—
ρ_m	Productivity penalty of mental illness	Homeless income over average	Meyer et al. (2025)	19.1%	—	—
λ	SUD hazard per unit of use	SUD rate	NSDUH 2024, ages 26+	3.68%	—	—
$\kappa_{\phi d}$	Mortality multiplier, SUD	Drug price participation elasticity	Saffer and Chaloupka (1999)	−0.63	—	—
μ_H	Recovery multiplier, homeless	Drug use disorder share of the homeless	Gutwinski et al. (2021)	21.7%	—	—
p_m	Prevalence of serious mental illness	Share of adults with serious mental illness	NSDUH 2023	5.5%	—	—
ζ_{DI}	Disability coverage given mental illness	Mental illness share of the homeless	Gutwinski et al. (2021)	25%	—	—
ω	Weight on the child	Fall in drug use with a child present	NSDUH 2024, reweighted	15.5%	—	—
δ_d	Transmission penalty, parental SUD	Child's SUD rate against a clean parent's	McGovern et al. (2023)	2 to 4	—	—
δ_h	Transmission penalty, parental homelessness	Set, not calibrated	Chetty, Hendren and Katz (2016)	0.31	—	—

3.1 Validation, and what is left undisciplined

None of the moments below is fitted. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught.

Moment	Data	Source
Homelessness rate by age, 25 to 34 through 65+	0.316 / 0.338 / 0.291 / 0.250 / 0.077 per cent	HUD PIT 2024
Shares of the homeless across those bands	26.0 / 27.2 / 21.0 / 18.4 / 7.5 per cent	HUD PIT 2024
Returns to homelessness within 24 months	17.4 per cent	HUD SPM 2024
Drug poisonings as a share of all deaths, 25 to 44	about 20 per cent	CDC WONDER
Bottom half's share of net worth	2.52 per cent	Fed DFA
Housing costs above 30 and above 50 per cent of income	40.6 and 17.5 per cent	SCF 2022
Age weighted standardised mortality ratio, SUD	2.5 to 9.8	Degenhardt et al.
Age weighted standardised mortality ratio, homeless	9.1 / 4.5 / 1.1 by band	Baggett et al.
Tenant contribution rate	31.2 per cent	HUD
Voucher holders with wages as their main income	28 per cent	HUD
Income with a moderate or severe SUD, over average	71.1 per cent	NSDUH 2024
Labour force participation, SUD against clean	65.8 against 61.5 per cent	NSDUH 2024
Homeless at treatment admission	14.5 per cent	TEDS-A 2022
Co-occurring psychiatric problem at admission	43.1 per cent	TEDS-A 2022
Working at all, among the homeless in treatment	6.6 per cent	TEDS-A 2022

Five parameters remain undisciplined, and they are flagged rather than hidden: the treatment effect on recovery g, the recovery multiplier for mental illness μ_m, the disability benefit level $\bar{y}_{DI}$, and the drug supply curvature ψ_2. Of these only g carries a headline result, and it is reported as a range rather than as a number.

One risk is larger than the others and belongs in the introduction rather than here. The best available quasi experimental evidence says the mortality benefit of these programmes runs through *housing*, and that conditioning on treatment adds nothing detectable. The submitted paper concluded that universal Treatment First dominates. The defensible position after this revision is the *generosity* result: expanding coverage dominates, and conditionality is second order and uncertain. That is consistent with the evidence and is still a paper, and getting there means reporting welfare and mortality as two dimensions, which is the AE's comment A9, rather than collapsing them into a single ranking.

What fixing permanent income costs, and what has to be refitted. The quantitative model holds permanent productivity fixed over the life cycle and keeps only the unemployment risk, which is what makes the child's penalty a shift along a single scalar rather than a shift across the nodes of an age varying discretisation. Two things then have to be redone rather than inherited. The dispersion of log productivity has to carry on its own what the fixed effect and the shock carried together, so it is set to the average cross sectional variance the stochastic process delivered rather than to its value at the youngest age. And the wealth moments are the ones at risk, since removing income risk removes precautionary saving: β is refitted to wealth over output, and the share of wealth held at the bottom of the distribution becomes an untargeted moment worth reporting rather than a target quietly still being hit.

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